Oracle DB

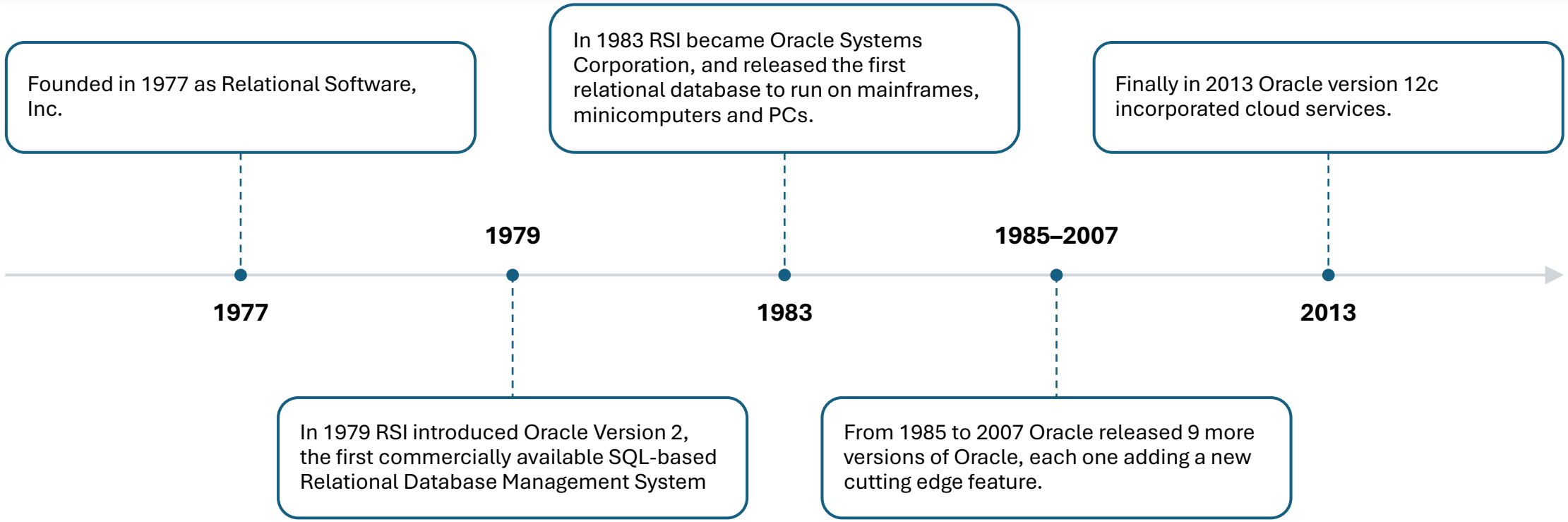
Darius Gillard

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Kevin Larson



History


Founded in 1977 as Relational Software, Inc.
1977
In 1979 RSI introduced Oracle Version 2, the first commercially available SQL-based Relational Database Management System
1979
In 1983 RSI became Oracle Systems Corporation, and released the first relational database to run on mainframes, minicomputers and PCs.
1983
From 1985 to 2007 Oracle released 9 more versions of Oracle, each one adding a new cutting edge feature.
1985–2007
Finally in 2013 Oracle version 12c incorporated cloud services.
2013

Cost

- Licensing fees, maintenance, and support contracts can **cost thousands to millions of dollars**, especially when compared to cheaper or even free alternatives like MySQL or PostgreSQL.
- The cost **increases sharply** as companies scale up their usage,add more features, or need enterprise-level support.

Industries that Use OracleDB



Data warehousing, and Enterprise Resource Planning



Telecommunications



Healthcare Systems



Financial Services

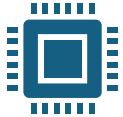
Companies that Use OracleDB

Netflix	LinkedIn	AirBnB	JPMorgan Chase
UnitedHealth Group	Zoom	Costco	Cisco

Why choose Oracle DB

- Oracle is best used for Large Scale, Complex applications with high performance needs.
- Oracle Database is the first database designed for enterprise grid computing, the most flexible and cost-effective way to manage information and applications. Enterprise grid computing creates large storage pools and industry-standard modular servers. With this architecture, each new system can be rapidly provisioned from the pool of components. There is no need for peak workloads, because capacity can be easily added or reassigned from resource pools as needed.

Oracle Editions



Express Edition (EX)

Free , caps on user size 12GB user data, 2GB RAM, Up 2 CPU threads



Enterprise Edition

Powerhouse design large enterprise and critical applications



Standard Edition

Popular among department-level teams and web applications, notably including oracle RAC (Real Application Cluster)



Standard Edition 2 (SE2)

The most cost-effective solution for medium businesses, offering many features but without RAC

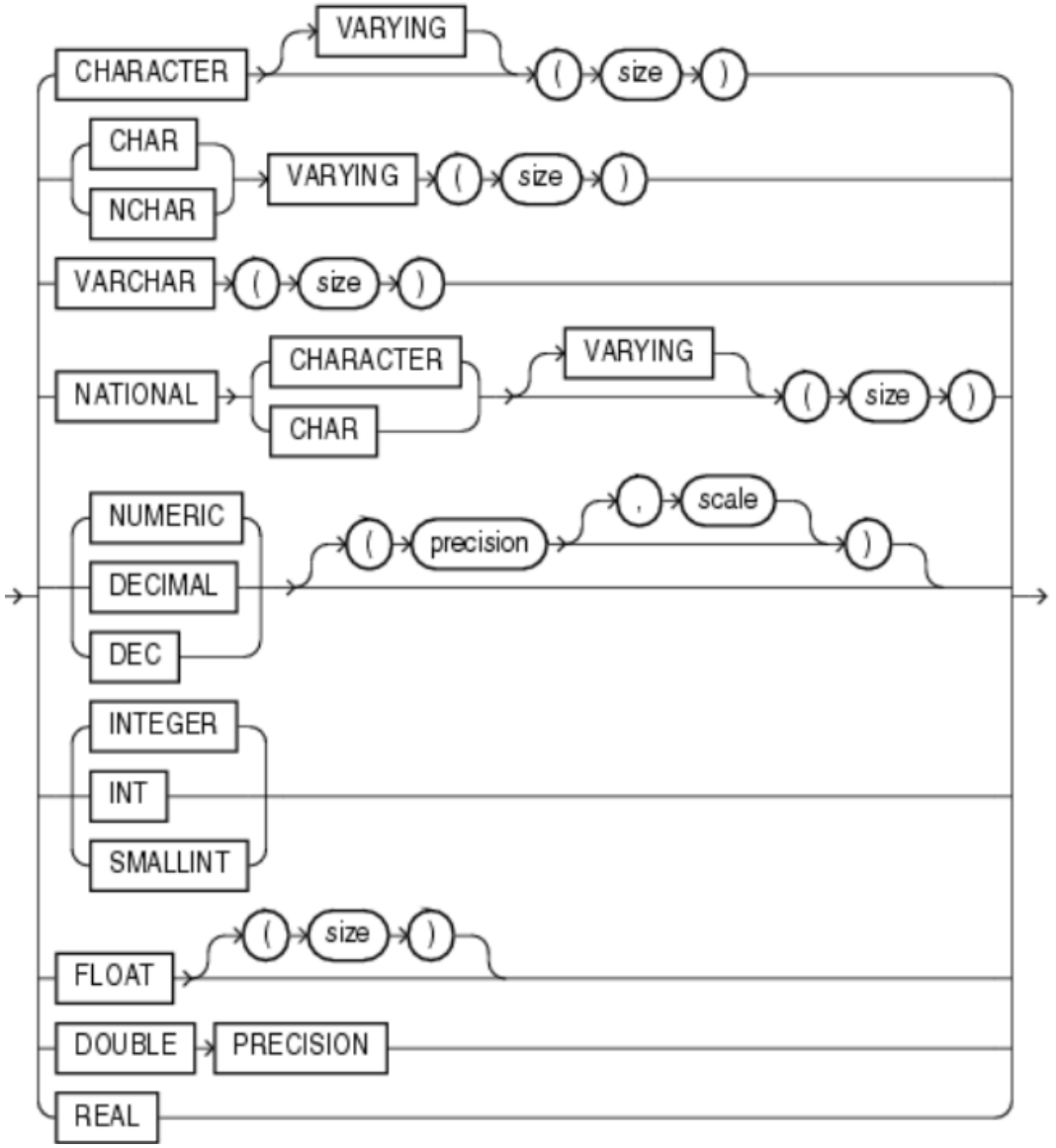


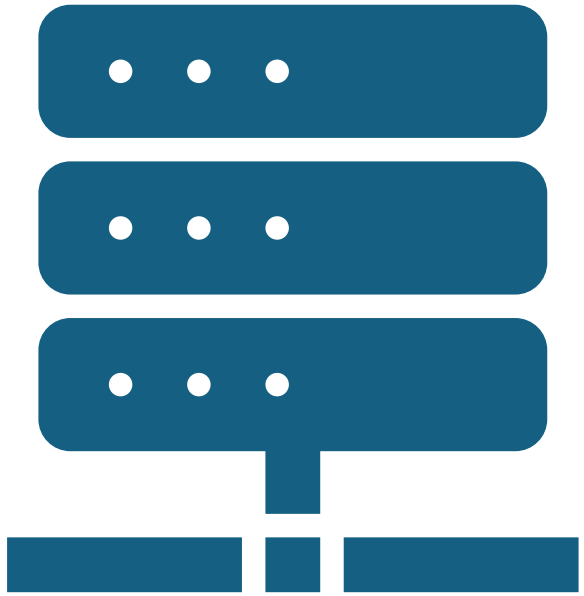
Personal Edition

This includes all the options and components that the Enterprise Edition for both development and deployment purposes.

Database Types

- Oracle Database supports many different data types:
  - XML is the standard way of identifying and describing data on the web.
  - LOB, this type of data allows you to store and manipulate large volumes of data.
  - Oracle Text, allows text search.
  - Oracle InterMedia, with services to develop and implement traditional applications.
  - Spatial Oracle, with functions to store and manage location content.





Architecture Principles

Reliability, Fault Tolerance, and High Availability

- A fundamental goal is ensuring that the database remains operational, and data is protected even when components fail

Scalability and Performance

- The architecture is designed to handle vast amounts of data and process requests concurrently with high performance, scaling to meet increasing demands

Portability, Flexibility, and Integration

- The ability to run on diverse platforms, connect to various data sources, support different deployment models, and integrate various data types and systems

Support

Oracle has many resources, including documentation, knowledge bases, and community forums that can be accessed on your own.

There is also a more specialized assistance through Oracle Premier Support or Oracle Advanced Customer Services for an additional premium.

Security

System Security: Controls access and use of the database at the system level**1** . This includes validating user name/password combinations (authentication), managing the amount of disk space and resource limits available to a user's schema objects, checking if a user is authorized to connect to the database, whether database auditing is active, and which system operations a user can perform

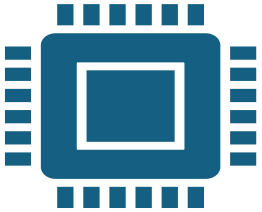
Data Security: Controls access and use of the database at the schema object level**7** . This includes determining which users have access to a specific schema object and the specific types of actions allowed for each user on that object

Redundancies

Oracle ensures various methods to ensure high availability and data protection. This includes mirroring data across multiple instances and using replication to keep copies synchronized.

Important: this allows users to continue working even if one database fails.

Pros and Cons of Oracle



Pro

Portability – Oracle easily runs on more than 100 hardware platforms and 20 networking protocols

Backup and Recovery – Oracle makes it easy to recover data with storage spaces and archive mechanisms

High Performance – Oracle offers good speed with large databases

Market Presence – Oracle is by far the largest RDBMS Vendor



Con

Complexity – The user has to be well trained in Oracle Database.

Cost – The price of Oracle can increase exponentially compared to other Database Solutions

Not for Smaller Companies – This system is not meant for small or medium sized companies. It's most useful for larger databases.

Question # 1

Which of the following industries commonly use Orcale DB ?

A) Automotive Design
B) Telecommunications
C) Education
D) Hospitality

Telecommunications companies

(like mobile phone carriers, internet providers, etc.) handle **massive amounts of real-time data** — such as phone calls, texts, data usage, customer accounts, billing, and network activity.

Answer and Reasoning

Question # 2

What is a *con* of using Oracle Database?

A) High market presence
B) Easy for small companies to implement
C) Expensive compared to competitors

D) Excellent portability



Answer and Reasoning

Expensive compared to competitors

Oracle Database is known for being **very powerful** and **feature-rich**, but one major downside is its **high cost**.

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